EduGuard: Hybrid AI Safety Guardrail System for Educational Support Chatbots

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1. INTRODUCTION

Educational chatbots can assist students by responding to questions related to academic policies, course selection, study guidance, and university support services. Despite these benefits, LLM-based chatbots used in higher education may also create several risks, including inaccurate responses, misinformation, bias, academic integrity issues, and unsafe or inappropriate guidance [1]. These risks are especially important in educational settings, where students may depend on chatbot answers when making academic decisions or seeking support.

EduGuard proposes a hybrid AI safety guardrail system for educational support chatbots. The system reviews each student message before generating a normal response and classifies it into one of five categories: SAFE, UNSAFE, SENSITIVE, OUT_OF_SCOPE, or PROMPT_INJECTION. This classification allows the chatbot to answer suitable academic questions, refuse requests related to cheating or harm, redirect unrelated topics, escalate sensitive student wellbeing concerns, and block prompt injection attempts.

Recent studies on AI guardrails emphasize the importance of reducing hallucinations, unsafe actions, privacy risks, and prompt-based attacks in LLM applications [2]. In addition, RAG can improve the reliability of educational chatbots by grounding responses in a trusted knowledge base rather than relying only on the language model [3]. Therefore, EduGuard integrates rule-based filtering, a DistilBERT safety classifier, FAISS-based RAG, GPT response generation for safe academic queries, and admin monitoring.

2. PROPOSED SYSTEM

The proposed system is designed as a multi-stage safety pipeline. The process begins when a student enters a message through the Student Chat interface. This message is then passed to the EduGuard Runtime Pipeline, which includes a rule-based safety filter, a DistilBERT safety classifier, decision logic, and response routing.

The rule-based filter is used to identify clear high-risk cases, such as cheating requests, direct prompt injection attempts, and sensitive wellbeing concerns. If the message cannot be fully handled by the rules, it is passed to the DistilBERT classifier to predict the appropriate safety label. After that, the decision layer selects the correct action for the request. SAFE messages are routed to the RAG and GPT components, while UNSAFE, SENSITIVE, OUT_OF_SCOPE, and PROMPT_INJECTION messages are managed before the system generates a normal chatbot response.

Label	Meaning	System Action
SAFE	Academic or student-support question	Answer using RAG and GPT
UNSAFE	Cheating, plagiarism, or disallowed request	Refuse and redirect
SENSITIVE	Student safety or wellbeing concern	Support and escalate
OUT_OF_SCOPE	Not related to education support	Redirect
PROMPT_INJECTION	Attempt to bypass rules or reveal hidden instructions	Block and alert admin

Table I. EduGuard safety labels and actions

3. DATASET AND KNOWLEDGE BASE

EduGuard relies only on synthetic data. The dataset is written entirely in English, and it does not include real student records, private university information, or actual support conversations. The main guardrail dataset consists of 800 records, with an equal distribution across the five labels: 160 samples for each label. The dataset is divided into 560 training samples, 120 validation samples, and 120 test samples. In addition, a separate benchmark set of 100 samples is used to evaluate boundary cases and safety-critical scenarios.

The project also includes a synthetic educational knowledge base for the RAG component. This knowledge base contains 200 chunks from a fictional educational institution. It covers academic policies, academic integrity, student support services, examinations, registration, graduation procedures, and educational FAQs. This design represents how a university chatbot can depend on controlled institutional content to improve response quality and reduce unsupported or ungrounded answers [3], [4].

A synthetic data approach was chosen because this project is a bootcamp prototype. This choice reduces privacy risks that could result from using real student data. At the same time, it allows the team to test the system's classification behavior, refusal logic, prompt injection blocking, and RAG-based academic support.

4. METHODOLOGY

The methodology is structured to balance safety, usefulness, and cost. To reduce unnecessary use of expensive components, the system applies low-cost safety checks before moving to response generation. The workflow starts with the student message, followed by the rule-based safety filter, the DistilBERT classifier, the decision layer, FAISS retrieval for SAFE requests, GPT response generation, and admin logging.

RAG is used only when a SAFE request requires academic or student-support information from the knowledge base. This helps improve reliability because the response is based on controlled and trusted content. Requests that are not classified as SAFE are handled before they reach GPT. UNSAFE requests are refused, SENSITIVE requests receive a supportive response and escalation, OUT_OF_SCOPE requests are redirected, and PROMPT_INJECTION requests are blocked and recorded.

The Admin Dashboard is added as a monitoring layer. It displays the predicted label, severity level, confidence score, matched rule, selected action, timestamp, and the student message. This improves transparency by showing what the system detected, why the decision was made, and what action was applied. Layered guardrail systems and monitoring are important because different LLM risks require different types of defense mechanisms [2], [5].

5. EVALUATION PLAN

The system is evaluated using the synthetic benchmark set and selected pipeline test cases. The evaluation focuses on overall accuracy, recall for each label, benchmark errors, over-refusal of SAFE questions, prevention of unsafe compliance, escalation of sensitive cases, prompt-injection blocking, and latency. These metrics are important because educational chatbots need to remain safe while still being useful and not overly restrictive.

In a preliminary prototype test, the benchmark accuracy reached 0.95, with 5 errors out of 100 samples. The safety-critical categories, including UNSAFE, SENSITIVE, OUT_OF_SCOPE, and PROMPT_INJECTION, achieved full recall on the benchmark set. Most of the remaining errors were SAFE questions that the system classified too cautiously. This result is acceptable for a safety-first prototype, although reducing over-refusal is still an important area for future improvement.

The final demonstration will use sample student questions from each label to test the complete workflow. These examples include a safe academic question, an academic dishonesty request, a wellbeing concern, an out-of-scope request, and a prompt injection attempt. The Admin Dashboard will also be used to display the logged decision and the action taken for each case.

6. COST AND ETHICS

EduGuard is designed with cost control as an important consideration. The system applies the rule-based filter and the DistilBERT classifier before calling GPT. As a result, GPT is used only for SAFE academic questions, while unsafe, sensitive, out-of-scope, and prompt-injection requests are handled earlier in the pipeline. This reduces unnecessary API usage and lowers the chance of generating risky or unsuitable responses.

From an ethical perspective, EduGuard focuses on protecting students, supporting academic integrity, and encouraging responsible AI use. The system avoids assisting with cheating or academic misconduct, provides supportive routing for sensitive wellbeing concerns, and allows admin review for high-risk cases. Since LLM chatbots in higher education may create risks related to misinformation, safety, and academic integrity, EduGuard uses classification and grounded responses to help reduce these risks [1], [2].

7. DELIVERABLES

The project deliverables include the Student App, the Admin Dashboard, the EduGuard runtime pipeline, the DistilBERT classifier, the FAISS RAG component, the synthetic benchmark dataset, documentation, and a demo video. The Student App shows how a learner interacts with the chatbot, while the Admin Dashboard shows how safety-related cases can be reviewed internally.

This proposal focuses on the system design and evaluation plan. The final report and presentation will include more detailed screenshots, implementation notes, and demonstration results.

8. LIMITATIONS AND FUTURE WORK

EduGuard is a bootcamp prototype and is not intended to be a production-ready system. The dataset is synthetic and English only, and the knowledge base is limited to a fictional educational institution. Some SAFE questions may be over-refused because the system follows a safety-first approach. In addition, the project runs locally using Streamlit and has not been deployed as a production API service using FastAPI.

Future work can improve EduGuard by expanding the dataset, adding human annotation, testing multilingual queries, improving RAG retrieval, and applying stronger red-team testing. Future versions can also explore stronger prompt-injection defenses and instruction hierarchy techniques to reduce the risk of user prompts overriding system-level rules [6].

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